25901429 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCPES200 - SCIENCE OF SPORTS TRAINING

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Power is defined as: [U] / [CO3]

   a). Maximum force only b). Work done per unit time

   c). Endurance capacity d). Flexibility level

2. Which is a type of speed? [An] / [CO4]

   a). Flexibility speed b). Strength speed

   c). Acceleration speed d). Balance speed

3. Periodization helps prevent: [A] / [CO6]

   a). Improved performance b). Overtraining and plateaus

   c). Skill development d). Strength gains

4. For adolescents, priority should be given to: [C] / [CO6]

   a). Winning competitions b). Long-term athletic development

   c). Immediate results d). Parental expectations

5. Exercise physiology helps in determining: [U] / [CO1]

a). Fan engagement b). Match venue

c). Team logo color d). Training intensity and load

6. Ballistic stretching involves: [An] / [CO4]

a). Holding stretches b). Slow movements

c). Bouncing movements d). No movement

7. Plyometric intensity can be modified by: [C] / [CO6]

a). Only adding weight b). Box height, distance, and body position

c). Only increasing reps d). No modification possible

8. Chatbots in sports coaching are mainly used for: [U] / [CO7]

a). Nutrition cooking b). Athlete communication support

c). Equipment repair d). Physical training demonstration only

9. For beginners, recommended frequency is: [C] / [CO6]

a). 7 days per week b). 2-3 days per week

c). Once per month d). Only when motivated

10. A gyroscope sensor in sports wearables is mainly used to measure: [U] / [CO7]

a). Blood sugar level b). Rotational movement and body orientation

c). VO2 max directly d). Skin color change

[1x10 = 10]

Part B

Short Answer

Answer any **5** questions

Each question carries **2** marks

11. What is training intensity? [U] / [CO2]

12. When to avoid static stretching? [An] / [CO4]

13. Give two agility drills. [C] / [CO6]

14. How does data analytics assist in injury prevention? [U] / [CO7]

15. Explain how resistance training improves muscular strength based on scientific principles. [U] / [CO2]

16. Sport-specific considerations? [C] / [CO6]

17. What is an electronic lap counter? [U] / [CO7]

[2x5 = 10]

Part C

Short Essay

Answer any **4** questions

Each question carries **5** marks

18. Describe the different types of training load and their significance in athletic preparation. [K] / [CO1]

19. Explain the strength-endurance continuum and its implications for training. [U] / [CO3]

20. Explain the phases of the stretch-shortening cycle and their role in enhancing muscular power. [U] / [CO6]

21. Describe the importance of hydration monitoring using biometric wearables. [U] / [CO7]

22. Describe how recovery science has advanced training practices. [U] / [CO2]

23. Design a flexibility program for a gymnast addressing different stretching types. [An] / [CO4]

[5x4 = 20]

Part D

Essay

Answer any **2** questions

Each question carries **15** marks

24. Explain the supercompensation theory in detail and its practical applications in periodized training programs. [K] / [CO1]

25. Discuss the factors affecting strength development and their consideration in personalized training. [U] / [CO3]

26. Design comprehensive plyometric programs for upper and lower body development. [C] / [CO6]

27. How would you apply Beep test results to improve cardiovascular endurance in athletes? [A] / [CO7]

[15x2 = 30]